

EXOhSPEC Stage-2 Thermal Diagnostics: V19, V20, V21

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Three passive monitor-only runs investigating the origin of the repeatable ~33-min thermal/spectral oscillation at the EXOhSPEC camera. **V19** (5 h) established the LK220-mediated cooling-chain hypothesis. **V20** (2 h) added a fourth PT104 probe directly on the camera rear (between the two coolant connections) with fully-confirmed LK220 telemetry. **V21** (2 h) repeated V20 with the camera's own CMOS TEC switched OFF, LK220 left ON, to isolate the camera's cooling duty cycle as a load modifier versus the oscillation's true origin. All statistics below are recomputed independently from the raw CSVs.

V19 — 5-hour passive baseline

Aim. Investigate why a strong AC-related oscillation appears in the cooling/camera temperatures while the internal instrument environment stays comparatively stable. **Dataset.** 301 frames over 5.00 h; 3612 LK220 samples at ~5 s cadence; bench TEC fixed at 22°C, no AO correction.

Bench TEC 22.001±0.012°C. Internal ECU 23.586±0.008°C (0.034°C p-p — very stable). BME 23.623±0.145°C (includes initial settling). PT104 AC 22.642±0.141°C, 0.535°C p-p, clearly periodic. Camera left/right/top ran **+1.34, +0.86, +1.72°C** above ECU (**+2.29, +1.81, +2.67°C** above the AC probe) — the camera is locally hotter than its surroundings and its oscillation is far larger than what the internal ECU records, so uniform box-air heating cannot explain it alone.

Source/channel	Mean	σ	P-P	r(L)	r(R)	r(Top)
Bench TEC	22.001°C	0.012	0.073	—	—	—
Internal ECU	23.586°C	0.008	0.034	0.104	0.154	0.119
BME/HVAC T	23.623°C	0.145	1.370*	0.089	0.067	0.143
PT104 AC probe	22.642°C	0.141	0.535	0.345	0.449	0.459
Camera left	24.928°C	0.104	0.396	—	—	—
Camera right	24.450°C	0.075	0.314	—	—	—
Camera top	25.310°C	0.133	0.530	—	—	—
LK220 TACT	21.993°C	0.149	0.490	0.896	0.902	0.907
LK220 TEC current	4.00 A	0.93	3.30	0.367	0.474	0.504
CoolerPower	72.58%	0.55	2.00	0.873	0.842	0.854

*BME peak-to-peak includes initial settling. dX_RMS = 0.076 px, dY_RMS = 0.537 px, dY within ±0.5 px = 54.2%, dY within ±0.3 px = 34.2%, OPL vs dY r = -0.45. Common period ≈ 33.4 min across AC, LK220 TACT/current, camera temps, CoolerPower, OPL and dY.

Working interpretation

AC oscillation → LK220 thermal response → coolant/camera heat sink → camera cooler response → camera-body temperature → optical/spectral drift, rather than uniform box-air heating.

Next step that motivated V20/V21

Place a PT104 probe directly on the camera heat sink beside the coolant connections, to test whether the sequence becomes LK220 TACT → heat sink → camera L/R/top → OPL/dY.

V19 — diagnostic plots

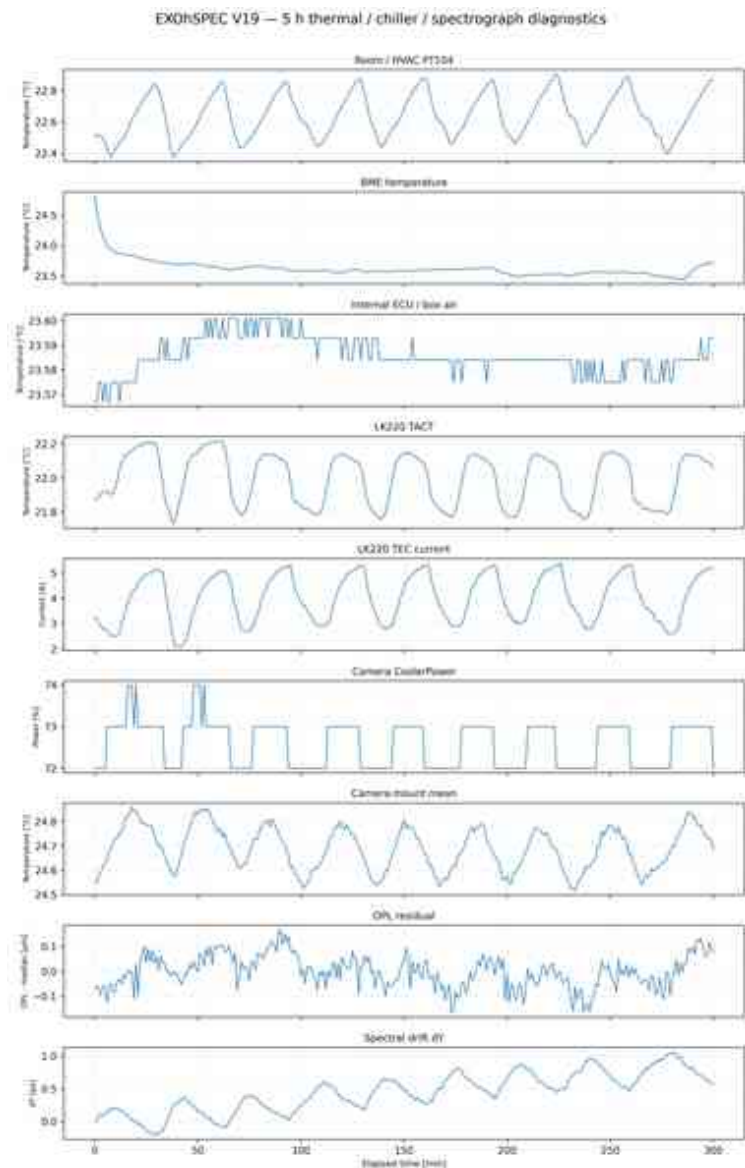


Figure V19.1. Nine-channel full-run comparison. The repeated cycle is visible in AC, LK220 TACT/current, CoolerPower, camera temperatures and dY, while the internal ECU stays comparatively stable.

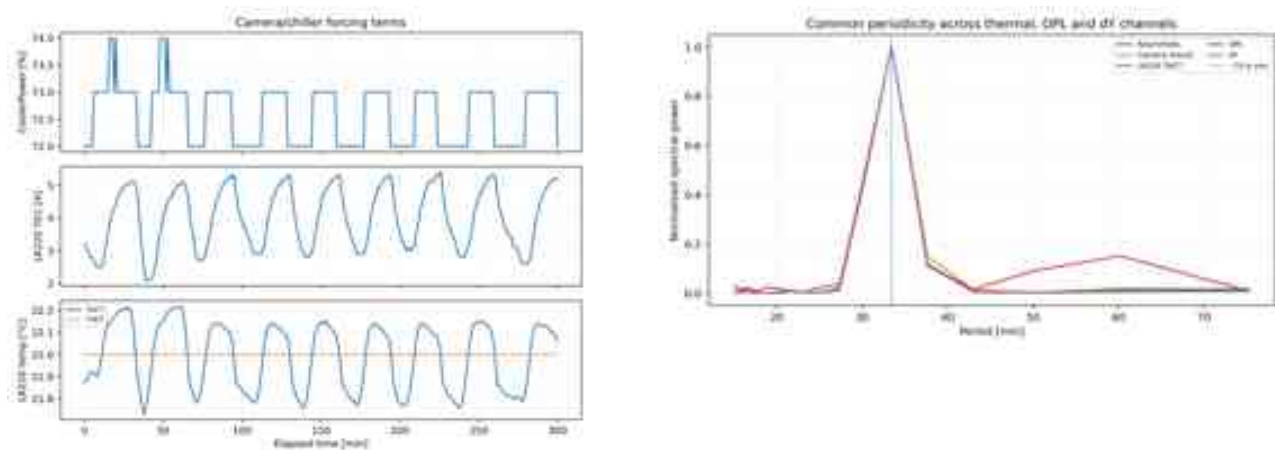


Figure V19.2 (left). CoolerPower, LK220 TEC current, LK220 TACT. **Figure V19.3 (right).** Detrended periodicity comparison, dominant common component near 33.4 min.

V20 — 2-hour repeat, confirmed LK220 telemetry, rear probe added

Aim. Repeat V19 conditions (camera TEC ON at −10°C) with fully-parsed LK220 MONITOR? telemetry and PT104 Ch4 relocated to the **camera rear, between the two coolant connections** (Bill's suggestion), to directly test the missing coolant–camera heat-sink link. **Dataset.** 121 frames over 2.00 h; 1452 LK220 samples at ~5 s.

Bench TEC 21.9997±0.0118°C. ECU 23.606±0.025°C. PT104 AC 22.612±0.152°C (0.612°C p-p). Camera left 25.455±0.135°C, right 24.392±0.080°C, **rear (new) 23.115±0.131°C**. Rear sits **−0.49°C** relative to ECU (cooler than ambient — it sits right at the coolant lines) while left/right remain **+1.85/+0.79°C** hotter: a clean spatial gradient coolant → heat sink → camera body. LK220 TACT 22.044±0.139°C (0.390°C p-p), TEC current 3.978±0.930 A (2.3–5.2 A), TEC duty 31.82±5.25% (new this run). CoolerPower 72.64±0.56% — essentially identical to V19's 72.58%, a good night-to-night consistency check.

Source/channel	Mean	σ	P-P	r(L)	r(R)	r(Rear)
Bench TEC	21.9997°C	0.0118	0.055	–	–	–
PT104 AC probe	22.612°C	0.152	0.612	−0.021	0.081	−0.116
BME/HVAC T	22.926°C	0.146	1.100*	−0.170	−0.181	−0.166
Camera left	25.455°C	0.135	0.481	–	–	–
Camera right	24.392°C	0.080	0.344	–	–	–
Camera rear (new)	23.115°C	0.131	0.484	–	–	–
LK220 TACT	22.044°C	0.139	0.390	0.667	0.690	0.611
LK220 TEC duty (new)	31.82%	5.25	16.2	see text	see text	see text
CoolerPower	72.64%	0.56	2.00	0.844	0.806	0.814

*BME p-p includes settling, as in V19. dX_RMS = 0.039 px, dY_RMS = 0.568 px, dY within ±0.5 px = 49.6%, within ±0.3 px = 18.2%, OPL vs dY r = 0.664. AC-vs-camera correlations flipped sign relative to V19 (there +0.35–0.46); with only ~2.6–3.1 cycles of the 33.4-min oscillation in 2 h, zero-lag Pearson r on a quasi-periodic signal is highly sensitive to the window's phase — read as a short-duration sampling artefact, not a changed physical link. CoolerPower and LK220 TACT stay the robust predictors in both runs.

Lag structure (±15 min window, to avoid locking onto an adjacent 33-min cycle): AC leads LK220 TACT by ~3 min (r=0.82). Rear leads camera-left by only ~1 min (r=0.99); camera-left leads CoolerPower by ~1 min (r=0.85) — both effectively simultaneous. The LK220–rear step itself was ambiguous at large lag; only the zero-lag value (r=0.61) is reported for that step.

Main result

The rear probe closes the V19 gap: it sits within 0.5°C of the coolant and leads camera-left by ~1 min (r=0.99). CoolerPower remains the single strongest, most reproducible correlate of camera-body temperature across V19 and V20 (r=0.81–0.87).

Open question carried into V21

V20 alone cannot separate “TEC duty causes the heat excess” from “TEC duty merely correlates with it because both respond to the same LK220/AC cycle.” V21 switches the camera TEC off to test this directly.

V20 — diagnostic plots

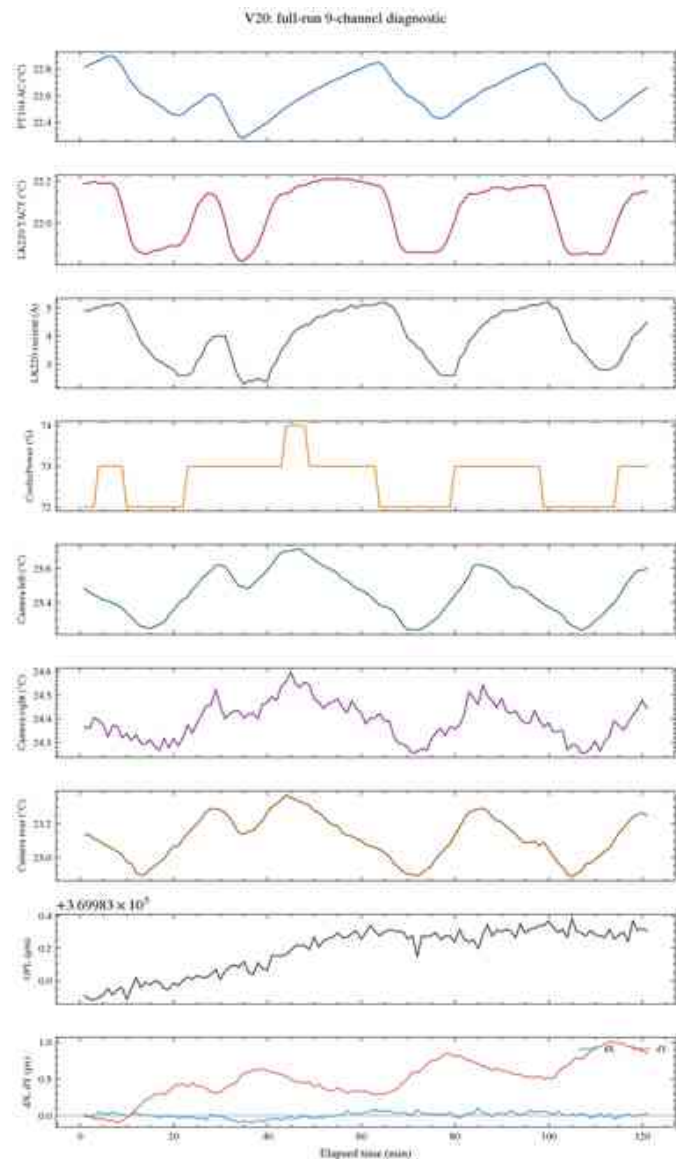


Figure V20.1. Nine-channel comparison; bottom panel shows dX and dY together per your request.

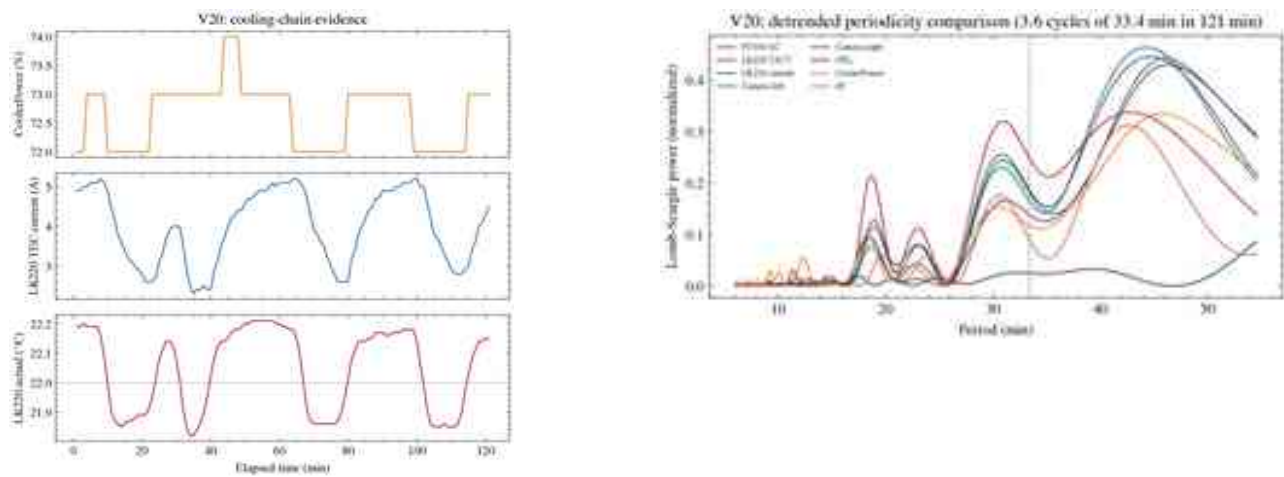


Figure V20.2 (left). CoolerPower, LK220 current, LK220 TACT. Figure V20.3 (right). Periodicity comparison (only ~2.6–3.1 cycles fit in this 2 h run, so treat the exact peak as indicative).

V21 — camera TEC OFF, LK220 still ON (controlled repeat of V20)

Aim. Repeat V20 with the camera's CMOS TEC switched **OFF**, LK220 left ON, to isolate whether the camera's own cooling duty is the *origin* of the oscillation or a downstream responder. **Dataset.** 121 frames, 2.00 h; CoolerOn=False, CoolerPower=0% for the entire run.

Thermal-step, not a steady baseline — frame 0 was taken right after the TEC-off switch. Fit $T(t)=T_{eq}-A\cdot e^{-t/\tau}$ to CMOS: **$T_{eq}=24.02\pm0.02^{\circ}\text{C}$, $A=17.12\pm0.35^{\circ}\text{C}$, $\tau=2.51\pm0.06\text{ min}$** . All steady-state numbers below use $t\geq 20\text{ min}$ ($\approx 8\tau$), applied identically to V20 for a fair comparison.

Post-20-min: camera left **+0.17°C** vs. ECU (was +1.85°C in V20), right **-0.21°C** (was +0.79°C), rear **-1.04°C** (was -0.49°C) — the local camera heat excess largely disappears with the TEC off.

Quantity	V20 (TEC ON)	V21 (TEC OFF)
CMOS temp	-9.99±0.04°C	24.02±0.02°C
CoolerPower	72.6±0.6%	0%
LK220 heat sink	31.39±1.84°C	29.40±1.74°C
LK220 TEC current	3.93±0.93 A	2.67±0.94 A
LK220 TEC duty	31.5±5.2%	24.3±5.3%
LK220 flow (confound check)	0.757 LPM	0.756 LPM
Camera rear	23.13±0.13°C	22.26±0.14°C
dY mean offset	0.582 px	1.110 px
dY std about own mean	0.203 px	0.224 px

Flow unchanged — rules out a flow-rate confound. Heat sink/current/duty all measurably lower with TEC off: the LK220's total load genuinely dropped. But dY's residual scatter about its own mean barely moves (0.203→0.224 px, ~10%) even though the mean offset nearly doubled (most plausibly a CMOS-package mechanical offset from running 34°C warmer, not a noise effect). **Turning the camera TEC off did not meaningfully quiet the oscillation.**

Most direct evidence in this dataset

With CoolerPower pinned at exactly 0% for the whole run, LK220's own TEC current still swings 1.0–4.9 A and TACT still oscillates ±0.2°C on the same ~32-min period. The camera cannot be driving this — it is doing nothing. The oscillation's origin is upstream, in the LK220/room-AC loop, now demonstrated under a condition where the camera is provably not the source.

Updated interpretation & caveats

AC/room → LK220 (self-sustaining, camera-independent) → coolant → camera body → OPL/dY. Camera-TEC duty scales the *magnitude* of local heat at the mount; the ~33-min *clock* lives upstream.

Caveats: V20/V21 ran on different nights (ECU 23.61 vs 23.32°C) — not a perfectly isolated single-variable swap. Several lags hit the ±15 min search boundary and are excluded as ambiguous. Correlation and shared periods do not by themselves prove causality.

V21 — diagnostic plots

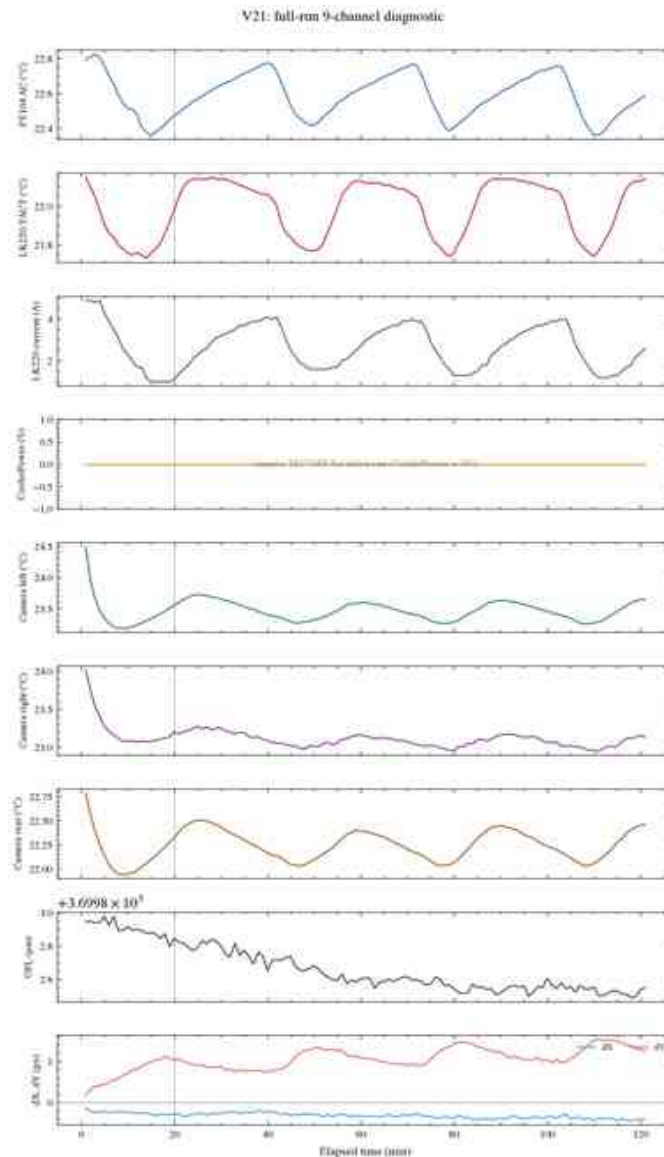


Figure V21.1. Dotted line at $t=20$ min marks the defensible steady-state start from the warm-up fit. CoolerPower panel is flat at 0 W — not a plotting artefact.

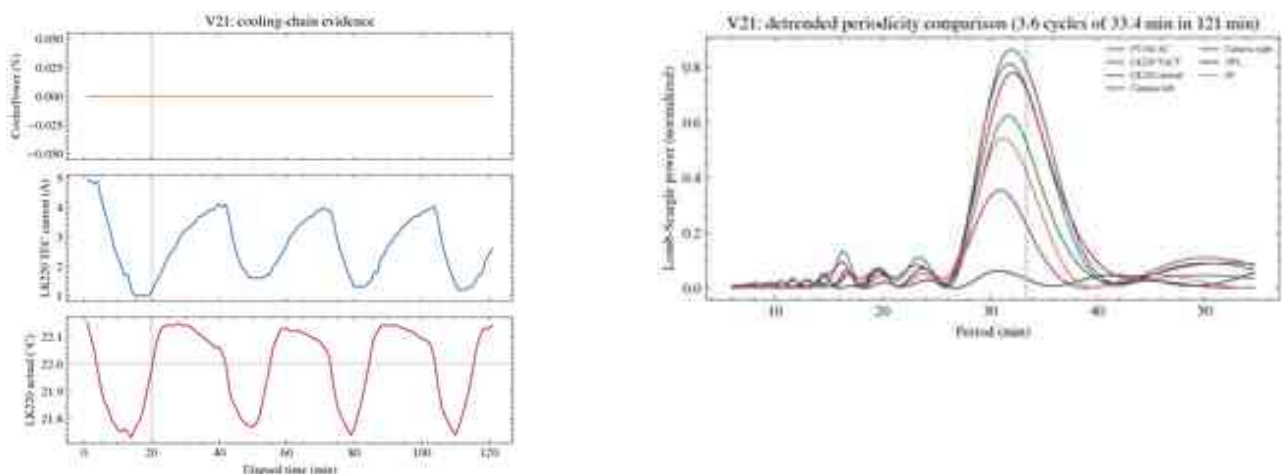


Figure V21.2 (left). CoolerPower flat at zero while LK220 current/TACT keep oscillating — direct evidence the oscillation originates upstream of the camera. Figure V21.3 (right). Periodicity comparison, CoolerPower omitted (constant zero has no spectrum); ~ 3.6 – 3.9 cycles span this 2 h run.

Next decisive measurement

V21 shows the LK220/room loop drives the oscillation independent of the camera. The decisive follow-up moves one level further upstream: attach the LK220's external TSP-TH sensor near the AC intake / room air (SENS=1, external control sensor per the LK220 manual) so the chiller regulates against the load point rather than its own internal return-line thermistor, and repeat. In parallel, run a full 5-hour TEC-off repeat matching V19's duration, so the ~ 33 -min period is resolved over ~ 9 cycles instead of the current ~ 3.6 – 3.9 .